

JKBOSE 2026

CLASS 11TH

**TRIGONOMETRIC
FUNCTIONS**

MATHS

LECTURE-06

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TOPICS TO BE COVERED

1

Exercise Discussion 3.3

$$* \cos x + \cos y = 2 \cos\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) \checkmark$$

$$* \cos x - \cos y = -2 \sin\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right) \checkmark$$

$$* \sin x + \sin y = 2 \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) \checkmark$$

$$* \sin x - \sin y = 2 \cos\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right) \checkmark$$

$$* \cos\left(\frac{\pi}{2} - x\right) = \cos\left(\frac{\pi}{2}\right)\cos x + \sin\left(\frac{\pi}{2}\right)\sin x$$

$$\Rightarrow \cos\left(\frac{\pi}{2} - x\right) = 0 \times \cos x + 1 \times \sin x$$

$$\Rightarrow \boxed{\cos\left(\frac{\pi}{2} - x\right) = \sin x}$$

$$* \boxed{\sin\left(\frac{\pi}{2} - x\right) = \cos x}$$

$$* \sin(\pi - x) = \sin x$$

$$* \cos(\pi - x) = -\cos x$$

$$* \boxed{\cos\left(\frac{\pi}{2} + x\right) = -\sin x}$$

$$* \boxed{\sin\left(\frac{\pi}{2} + x\right) = \cos x}$$

$$* \boxed{\sin(\pi + x) = -\sin x}$$

$$* \boxed{\cos(\pi + x) = -\cos x}$$

$$* \operatorname{cosec}(\pi + x) = \frac{1}{\sin(\pi + x)} = \frac{1}{-\sin x} = -\operatorname{cosec} x$$

EXERCISE – 3.3

QUESTION

Prove that

$$\sin^2 \frac{\pi}{6} + \cos^2 \frac{\pi}{3} - \tan^2 \frac{\pi}{4} = -\frac{1}{2}$$

Proof: $\text{LHS} = \sin^2\left(\frac{\pi}{6}\right) + \cos^2\left(\frac{\pi}{3}\right) - \tan^2\left(\frac{\pi}{4}\right)$

$$\Rightarrow \text{LHS} = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 - 1^2$$

$$\Rightarrow \text{LHS} = \frac{1}{4} + \frac{1}{4} - 1 = \frac{1}{2} - 1 = -\frac{1}{2}$$

$$\Rightarrow \text{LHS} = \text{RHS.}$$

QUESTION

Prove that

$$2\sin^2\frac{\pi}{6} + \operatorname{cosec}^2\frac{7\pi}{6} \cos^2\frac{\pi}{3} = \frac{3}{2}$$

Proof: $\text{LHS} = 2\sin^2\left(\frac{\pi}{6}\right) + \operatorname{cosec}^2\left(\frac{7\pi}{6}\right) \cos^2\left(\frac{\pi}{3}\right)$

$$\Rightarrow \text{LHS} = 2 \times \left(\frac{1}{2}\right)^2 + \left[\operatorname{cosec}\left(\pi + \frac{\pi}{6}\right)\right]^2 \times \left(\frac{1}{2}\right)^2$$

$$\Rightarrow \text{LHS} = \cancel{2} \times \frac{1}{\cancel{4}} + \left[-\operatorname{cosec}\left(\frac{\pi}{6}\right)\right]^2 \times \frac{1}{4}$$

$$\Rightarrow \text{LHS} = \frac{1}{2} + (-2)^2 \times \frac{1}{4}$$

$$\Rightarrow \text{LHS} = \frac{1}{2} + \cancel{4} \times \frac{1}{\cancel{4}}$$

$$\Rightarrow \text{LHS} = \frac{1}{2} + 1 = \frac{3}{2}$$

$$\Rightarrow \boxed{\text{LHS} = \text{RHS}}$$

QUESTION



Prove that

$$\cot^2 \frac{\pi}{6} + \operatorname{cosec} \frac{5\pi}{6} + 3 \tan^2 \frac{\pi}{6} = 6$$

Proof: $LHS = \cot^2\left(\frac{\pi}{6}\right) + \operatorname{cosec}\left(\frac{5\pi}{6}\right) + 3 \tan^2\left(\frac{\pi}{6}\right)$

$$\Rightarrow LHS = \left(\frac{1}{\sqrt{3}}\right)^2 + \operatorname{cosec}\left(\pi - \frac{\pi}{6}\right) + 3 \times \left(\frac{1}{\sqrt{3}}\right)^2$$
$$\Rightarrow LHS = 3 + \operatorname{cosec}\left(\frac{\pi}{6}\right) + \cancel{3} \times \frac{1}{\cancel{3}}$$
$$\Rightarrow LHS = 3 + 2 + 1 = 6 = \overline{RHS}$$
$$\Rightarrow \boxed{LHS = RHS.}$$

QUESTION

Prove that

$$2\sin^2\frac{3\pi}{4} + 2\cos^2\frac{\pi}{4} + 2\sec^2\frac{\pi}{3} = 10$$

Homework.

QUESTION

Find the value of:

(i) $\sin 75^\circ$

(ii) $\tan 15^\circ$

(Sol) (i) we have;

$$\sin 75^\circ = \sin(45^\circ + 30^\circ)$$

$$\Rightarrow \sin 75^\circ = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ$$

$$\Rightarrow \sin 75^\circ = \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2}$$

$$\Rightarrow \sin 15^\circ = \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} = \frac{\sqrt{3} + 1}{2\sqrt{2}}$$

(ii) we have

$$\tan 15^\circ = \tan(45^\circ - 30^\circ)$$

$$\Rightarrow \tan 15^\circ = \frac{\tan 45^\circ - \tan 30^\circ}{1 + \tan 45^\circ \tan 30^\circ} = \frac{\left(1 - \frac{1}{\sqrt{3}}\right)}{\left(1 + \frac{1}{\sqrt{3}}\right)}$$

$$\Rightarrow \tan 15^\circ = \frac{\left(\frac{\sqrt{3}-1}{\sqrt{3}}\right)}{\left(\frac{\sqrt{3}+1}{\sqrt{3}}\right)} = \frac{\sqrt{3}-1}{\sqrt{3}+1}$$

$$\Rightarrow \tan 15^\circ = \frac{\sqrt{3}-1}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1}$$

$$\Rightarrow \tan 15^\circ = \frac{(\sqrt{3}-1)^2}{\cancel{\sqrt{3}^2}-1^2} = \frac{\cancel{\sqrt{3}^2}+1^2-2\sqrt{3}}{2}$$

$$\Rightarrow \tan 15^\circ = \frac{4-2\sqrt{3}}{2} = \cancel{2} \frac{(2-\sqrt{3})}{\cancel{2}}$$

$$\Rightarrow \boxed{\tan 15^\circ = 2-\sqrt{3}}$$

QUESTION



Prove the following:

$$\cos\left(\frac{\pi}{4} - x\right) \cos\left(\frac{\pi}{4} - y\right) - \sin\left(\frac{\pi}{4} - x\right) \sin\left(\frac{\pi}{4} - y\right) = \sin(x + y)$$

Proof:

$$\begin{aligned} \text{LHS} &= \cos\left(\frac{\pi}{4} - x\right) \cos\left(\frac{\pi}{4} - y\right) - \sin\left(\frac{\pi}{4} - x\right) \sin\left(\frac{\pi}{4} - y\right) \\ &= \cos\left[\left(\frac{\pi}{4} - x\right) + \left(\frac{\pi}{4} - y\right)\right] \\ &= \cos\left[\frac{\pi}{2} - (x + y)\right] = \sin(x + y) = \text{RHS.} \end{aligned}$$

QUESTION

Prove the following:

$$\frac{\tan\left(\frac{\pi}{4} + x\right)}{\tan\left(\frac{\pi}{4} - x\right)} = \left(\frac{1 + \tan x}{1 - \tan x}\right)^2$$

Proof: We know that

$$\tan\left(\frac{\pi}{4} + x\right) = \frac{\tan\frac{\pi}{4} + \tan x}{1 - \tan\frac{\pi}{4}\tan x}$$

$$\Rightarrow \tan\left(\frac{\pi}{4} + x\right) = \frac{1 + \tan x}{1 - \tan x} \quad \text{--- (1)}$$

$$\text{Also, } \tan\left(\frac{\pi}{4} - x\right) = \frac{\tan\left(\frac{\pi}{4}\right) - \tan x}{1 + \tan\left(\frac{\pi}{4}\right)\tan x}$$

$$\Rightarrow \tan\left(\frac{\pi}{4} - x\right) = \frac{1 - \tan x}{1 + \tan x} \quad \text{--- (2)}$$

Dividing (1) and (2), we get,

$$\begin{aligned} \frac{\tan\left(\frac{\pi}{4} + x\right)}{\tan\left(\frac{\pi}{4} - x\right)} &= \frac{\left(\frac{1 + \tan x}{1 - \tan x}\right)}{\left(\frac{1 - \tan x}{1 + \tan x}\right)} = \frac{1 + \tan x}{1 - \tan x} \times \frac{1 + \tan x}{1 - \tan x} \\ &= \frac{(1 + \tan x)^2}{(1 - \tan x)^2} \end{aligned}$$

$$\Rightarrow \frac{\tan\left(\frac{\pi}{4} + x\right)}{\tan\left(\frac{\pi}{4} - x\right)} = \left(\frac{1 + \tan x}{1 - \tan x}\right)^2$$

Hence proved

QUESTION

Prove the following:

$$\frac{\cos(\pi + x)\cos(-x)}{\sin(\pi - x)\cos\left(\frac{\pi}{2} + x\right)} = \cot^2 x$$

$$\cos\left(\frac{\pi}{2} + x\right) = -\sin x$$

Proof: LHS =
$$\frac{\cos(\pi + x)\cos(-x)}{\sin(\pi - x)\cos\left(\frac{\pi}{2} + x\right)}$$

$$= \frac{-\cos x \times \cos x}{\sin x \times -\sin x} = \frac{\cos^2 x}{\sin^2 x} = \cot^2 x = \text{RHS}$$

QUESTION

Prove the following:

$$\cos\left(\frac{3\pi}{2} + x\right) \cos(2\pi + x) \left[\cot\left(\frac{3\pi}{2} - x\right) + \cot(2\pi + x) \right] = 1$$

Proof: LHS = $\cos\left(\frac{3\pi}{2} + x\right) \cos(2\pi + x) \left[\cot\left(\frac{3\pi}{2} - x\right) + \cot(2\pi + x) \right]$

$$\Rightarrow \text{LHS} = \sin x \cos x \left[\tan x + \cot x \right]$$

$$\Rightarrow \text{LHS} = \sin x \cos x \left[\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} \right] = \cancel{\sin x \cos x} \left(\frac{\sin^2 x + \cos^2 x}{\cancel{\sin x \cos x}} \right)$$

$$\Rightarrow \text{LHS} = 1 = \text{RHS}$$

$$\Rightarrow \boxed{\text{LHS} = \text{RHS}}$$

QUESTION



Prove the following:

$$\sin \frac{(n+1)x}{2} \sin \frac{(n+2)x}{2} + \cos \frac{(n+1)x}{2} \cos \frac{(n+2)x}{2} = \cos x$$

$$\cos(x-y)$$

Homework Problem.

QUESTION

Prove the following:

$$\cos\left(\frac{3\pi}{4} + x\right) - \cos\left(\frac{3\pi}{4} - x\right) = -\sqrt{2} \sin x$$

Proof: we know that

$$\cos(x+y) - \cos(x-y) = -2\sin x \sin y$$

$$\begin{aligned} \text{So } \cos\left(\frac{3\pi}{4} + x\right) - \cos\left(\frac{3\pi}{4} - x\right) &= -2\sin\left(\frac{3\pi}{4}\right) \sin x \\ &= -2\sin\left(\pi - \frac{\pi}{4}\right) \sin x \end{aligned}$$

$$= -2\sin\left(\frac{\pi}{4}\right) \sin x$$

$$= -2 \times \frac{1}{\sqrt{2}} \sin x = -\frac{2}{\sqrt{2}} \sin x$$

$$= -\frac{2}{\sqrt{2}} \sin x \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{-2\sqrt{2} \sin x}{2}$$

$$= -\sqrt{2} \sin x = \text{RHS.}$$

QUESTION

Prove the following:

$$\sin^2 6x - \sin^2 4x = \sin 2x \sin 10x$$

$$\begin{aligned} * \sin x + \sin y &= 2 \sin\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) \\ * \sin x - \sin y &= 2 \cos\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right) \end{aligned}$$



Proof: LHS = $\sin^2 6x - \sin^2 4x$

$$\Rightarrow \text{LHS} = (\sin 6x + \sin 4x)(\sin 6x - \sin 4x)$$

$$\Rightarrow \text{LHS} = 2 \sin\left(\frac{6x+4x}{2}\right) \cos\left(\frac{6x-4x}{2}\right) \times 2 \cos\left(\frac{6x+4x}{2}\right) \sin\left(\frac{6x-4x}{2}\right)$$

$$\Rightarrow \text{LHS} = 2 \sin 5x \cos x \times 2 \cos 5x \sin x.$$

$$\Rightarrow \text{LHS} = (2 \sin x \cos x)(2 \sin 5x \cos 5x) = \sin 2x \sin 10x = \text{RHS}.$$

QUESTION

Prove the following:

$$\cos^2 2x - \cos^2 6x = \sin 4x \sin 8x$$

Homework

$$\begin{aligned} * \cos x + \cos y &= 2 \cos\left(\frac{x+y}{2}\right) \cos\left(\frac{x-y}{2}\right) \\ * \cos x - \cos y &= -2 \sin\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right) \end{aligned}$$

QUESTION

Prove the following:

$$\sin 2x + 2 \sin 4x + \sin 6x = 4 \cos^2 x \sin 4x$$

Proof: LHS = $\sin 2x + 2 \sin 4x + \sin 6x$

$$\Rightarrow \text{LHS} = (\sin 2x + \sin 6x) + 2 \sin 4x$$

$$\Rightarrow \text{LHS} = 2 \sin \left(\frac{2x+6x}{2} \right) \cos \left(\frac{2x-6x}{2} \right) + 2 \sin 4x$$

$$\Rightarrow \text{LHS} = 2 \sin 4x \cos 2x + 2 \sin 4x$$

$$\Rightarrow \text{LHS} = 2 \sin 4x (\cos 2x + 1)$$

$$\Rightarrow \text{LHS} = 2 \sin 4x (2 \cos^2 x - 1 + 1)$$

$$\Rightarrow \text{LHS} = 4 \cos^2 x \sin 4x = \text{RHS}$$

QUESTION

Prove the following:

$$\cot 4x(\sin 5x + \sin 3x) = \cot x (\sin 5x - \sin 3x)$$

Proof: LHS = $\cot 4x (\sin 5x + \sin 3x)$

$$\Rightarrow \text{LHS} = \frac{\cos 4x}{\sin 4x} \times 2 \sin\left(\frac{5x+3x}{2}\right) \cos\left(\frac{5x-3x}{2}\right)$$

$$\Rightarrow \text{LHS} = \frac{\cos 4x}{\cancel{\sin 4x}} \times 2 \cancel{\sin 4x} \cos x$$

$$\Rightarrow \boxed{\text{LHS} = 2 \cos x \cos 4x} \text{---(1)}$$



$$\text{Also RHS} = \cot x (\sin 5x - \sin 3x)$$

$$\Rightarrow \text{RHS} = \frac{\cos x}{\sin x} \times 2 \cos\left(\frac{5x+3x}{2}\right) \sin\left(\frac{5x-3x}{2}\right)$$

$$\Rightarrow \text{RHS} = \frac{\cos x}{\cancel{\sin x}} \times 2 \cos 4x \cancel{\sin x}$$

$$\Rightarrow \boxed{\text{RHS} = 2 \cos x \cos 4x} \text{---(2)}$$

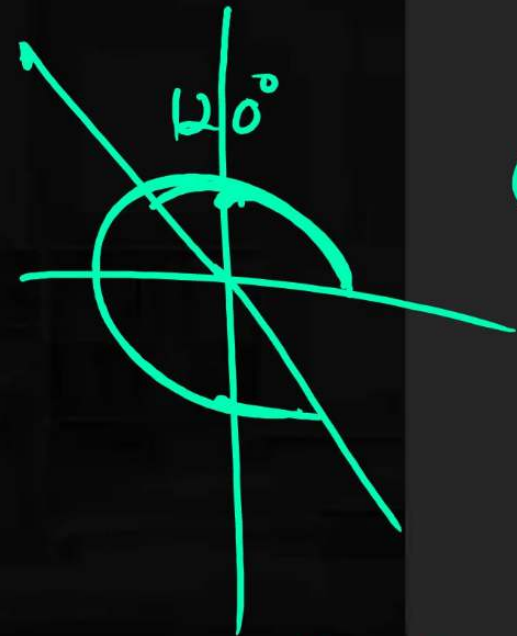
From (1) and (2), we get;

$$\boxed{\text{LHS} = \text{RHS}}$$

Hence proved.

$$\frac{5\pi}{6} = \pi - \frac{\pi}{6}$$

$$\frac{5}{6} \times 180^\circ = 150^\circ$$



$$90^\circ + 30^\circ = 120^\circ$$

$$\cos\left(\frac{\pi}{2} + \frac{\pi}{6}\right)$$

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$$270^\circ + 30^\circ$$

$$\frac{3\pi}{2} + \frac{\pi}{6}$$

Thank You